Project Title

IndicSearch — Fast Keyword-Based Search Engine for Indian-Language Wikipedia

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Data Structures and Algorithms -3 (25CS2103E)

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Submitted by
Ch Suraj Reddy (2520030355)
G Kevin Josh (2520030213)
Veeranala Chaitanaya (2520039619)
Team Number: 1
Section: 4

Under the guidance of

Mrs. J. Yamini Devi

Asst. Professor, CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

A Fast Keyword-Based Search Engine for Indian-Language Wikipedia is a search application designed to retrieve relevant Wikipedia articles written in Indian languages. The project aims to improve access to multilingual information, with a focus on Telugu and potential support for languages such as Hindi, Tamil, Kannada, Malayalam, Bengali, Marathi, Gujarati, and Punjabi. The system uses an inverted index implemented with hash tables and dictionaries to map keywords to the documents containing them. Sorted posting lists are used to represent document occurrences, while the two-pointer algorithm efficiently processes multiple keywords through intersection and union operations. These techniques help reduce unnecessary comparisons and improve query-processing efficiency compared with repeatedly scanning the entire document collection. Unicode normalization and text tokenization support the processing of Indian-language content, and TF-IDF ranking can be used to arrange retrieved articles according to their relevance to the user's query.

The project combines fundamental Data Structures and Algorithms concepts with web application development to demonstrate the practical implementation of information retrieval techniques. IndicSearch provides a foundation for multilingual keyword searching and can be extended with spelling correction, semantic search, improved relevance ranking, and support for additional Indian languages.

Keywords: IndicSearch, Data Structures and Algorithms, Inverted Index, Hash Tables, Two-Pointer Algorithm, Unicode, Information Retrieval, TF-IDF.

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